

```

within Buildings.Fluid.Chillers.Examples;
model ElectricReformulatedEIR
  "Test model for chiller electric reformulated EIR"
  extends Modelica.Icons.Example;
  extends Buildings.Fluid.Chillers.Examples.BaseClasses.PartialElectric(
    P_nominal=-per.QEva_flow_nominal/per.COP_nominal,
    mEva_flow_nominal=per.mEva_flow_nominal,
    mCon_flow_nominal=per.mCon_flow_nominal,
    soul(nPorts=1),
    sou2(nPorts=1));

  parameter
    Data.ElectricReformulatedEIR.ReformEIRChiller_McQuay_WSC_471kW_5_89COP_Vanes
    per "Chiller performance data"
    annotation (Placement(transformation(extent={{60,80},{80,100}})));

  Buildings.Fluid.Chillers.ElectricReformulatedEIR chi(
    redeclare package Medium1 = Medium1,
    redeclare package Medium2 = Medium2,
    per=per,
    energyDynamics=Modelica.Fluid.Types.Dynamics.FixedInitial,
    dp1_nominal=6000,
    dp2_nominal=6000) "Chiller model"
    annotation (Placement(transformation(extent={{0,0},{20,20}})));

equation
  connect(soul.ports[1], chi.port_a1) annotation (Line(
    points={{-40,16},{0,16}},
    color={0,127,255},
    smooth=Smooth.None));
  connect(chi.port_b1, res1.port_a) annotation (Line(
    points={{20,16},{26,16},{26,40},{32,40}},
    color={0,127,255},
    smooth=Smooth.None));
  connect(sou2.ports[1], chi.port_a2) annotation (Line(
    points={{40,4},{20,4}},
    color={0,127,255},
    smooth=Smooth.None));
  connect(chi.port_b2, res2.port_a) annotation (Line(
    points={{0,4},{-10,4},{-10,-20},{-20,-20}},
    color={0,127,255},
    smooth=Smooth.None));
  connect(chi.on, greaterThreshold.y) annotation (Line(
    points={{-2,13},{-8,13},{-8,90},{-19,90}},
    color={255,0,255},
    smooth=Smooth.None));
  connect(TSet.y, chi.TSet) annotation (Line(
    points={{-59,60},{-20,60},{-20,8},{-2,8},{-2,7}},
    color={0,0,127},
    smooth=Smooth.None));
  annotation (
    experiment(Tolerance=1e-6, StopTime=14400),
    __Dymola_Commands(file="modelica://Buildings/Resources/Scripts/Dymola/Fluid/Chillers/
    Examples/ElectricReformulatedEIR.mos"
      "Simulate and plot"), Documentation(info="<html>
    <p>
    Example that simulates a chiller whose efficiency is computed based on the
    condenser leaving and evaporator leaving fluid temperature.
    A bicubic polynomial is used to compute the chiller part load performance.
    </p>
    </html>", revisions="<html>
    <ul>
    <li>
    August 14, 2014, by Michael Wetter:<br/>
    Added missing <code>redeclare</code> keyword in
    performance data redeclaration.
    This avoids an error in OpenModelica.
    </li>
    <li>

```

```
September 17, 2010, by Michael Wetter:<br/>
First implementation.
</li>
</ul>
</html>"),
    Diagram(coordinateSystem(extent={{-100,-100},{100,120}})),
    Icon(coordinateSystem(extent={{-100,-100},{100,100}}));
end ElectricReformulatedEIR;
```